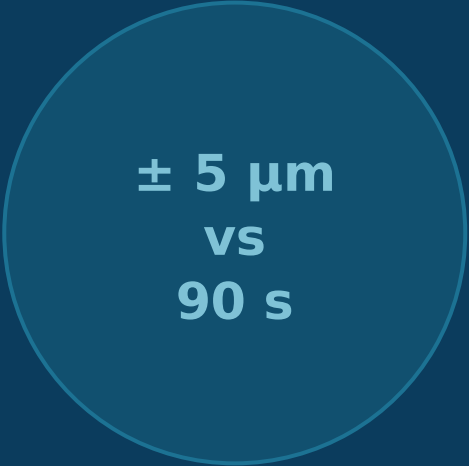


CASE STUDY & SEMINAR · AUTOMATION IN MANUFACTURING

# In-Line Quality Inspection Strategy



$\pm 5 \mu\text{m}$   
VS  
90 s

Comparative analysis of Coordinate Measuring Machines against non-contact vision and laser inspection, for an automated engine cylinder-block line

Target COs: CO-1, CO-5

SDG 9 — Industry, Innovation & Infrastructure

SDG 4 — Quality Education

# The problem: inspection has become the bottleneck

Automated cylinder-block line · blocks are currently carried off the line for offline dimensional checking

**90 s**

line takt time

**800**

blocks per day  
(two shifts)

**25 min**

to inspect one block  
on a CMM

**2.0 %**

present scrap  
and rework rate

## Two consequences

- Inspection is the throughput bottleneck — the line waits on the inspection room.
- Far more damaging: the DETECTION LAG. With sampling only, a drifting bore diameter is found long after a batch of blocks has already been machined to the wrong size.
- Scrap is therefore not a measurement problem but a feedback-speed problem.

# What this seminar covers

Three analyses, then a recommendation

1

## Tactile CMM

Structural configurations · drive control · probe systems · accuracy tests, laser-interferometric calibration and error compensation

2

## Non-contact inspection

3-D laser line scanners · CCD/CMOS machine vision · comparison across range, speed, accuracy, repeatability, reflectivity and environment

3

## Decision

Weighted trade-off matrix · throughput arithmetic · economics · final engineering recommendation

# Four structural configurations

Which structure suits which part — the trade-off between accuracy, stiffness, access and part size

## Moving bridge

*Bridge travels over a fixed granite table*

- + Best accuracy-to-cost; part mass does not enter the dynamics
- Bridge yaw if single-driven

## Fixed bridge

*Bridge fixed, the TABLE moves*

- + Highest accuracy — the metrology frame never moves
- Moving table limits part weight

## Cantilever

*Probe on an arm from one side*

- + Open on three sides — easy robot loading; small footprint
- Arm deflects with extension

## Gantry / Horizontal arm

*Self-supporting frame / horizontal quill*

- + Very large volumes; part craned in; good access to vertical faces
- Lower accuracy per metre; arm droop

**Heavy parts (a 38 kg block) → moving bridge.**

**Light parts → cantilever.**

**Very large parts → gantry.**

# Drive control and probe system

Both selections are forced by the need to run unattended inside a production cell

## Drive control

### Manual (free-floating)

operator pushes the carriages — repeatability depends on the operator. Rejected.

### Motorised / joystick

consistent approach speed, but still operator-paced. Teaching only.

### Direct Computer Control

a part program drives the probe — repeatable, unattended, line-triggerable, and the only option that can scan. SELECTED.

## Probe system

### Touch-trigger

3–5 points per second. Sizes and positions only. Main error is pre-travel variation — the trigger force changes with approach direction.

### Continuous scanning

Hundreds to thousands of points per second. Measures FORM — roundness, cylindricity, profile. SELECTED.

***Cylinder bores need cylindricity — and that cannot be certified from a handful of discrete points.***

# Accuracy, calibration and traceability

A CMM is itself a measuring instrument — it must be calibrated and periodically reverified

**3**

linear positioning  
(scale) errors

**6**

straightness  
errors

**9**

angular errors  
(pitch, yaw, roll)

**3**

squareness errors  
between axis pairs

**21**

rigid-body error  
components in total

## How they are measured

- Laser interferometry — linear positioning directly; pitch and yaw with angular optics; straightness with Wollaston optics
- Roll by electronic level or autocollimator
- Squareness by granite square, ball plate or ball bar
- Air temperature, pressure and humidity fed in to correct the refractive index

## Compensation and acceptance

- The 21 components are loaded as a CAA volumetric compensation map — every point corrected in real time, improving volumetric accuracy 3-10×
- ISO 10360-2 (IS 15250): length error  $E_0$ , MPE and repeatability  $R_0$
- ISO 10360-5: probing form error P on a calibrated sphere
- ISO 10360-4: scanning probing error THP

# Two non-contact technologies

Both trade absolute accuracy for speed and coverage

## 3-D laser line scanner

*A laser line is projected on the part and imaged by a camera at a known angle; triangulation gives 3-D coordinates.*

- 10 000 – 600 000 points per second — a whole block in 2–5 min
- Full-field point cloud → colour deviation map against CAD
- 20–50  $\mu\text{m}$  per point; 10–30  $\mu\text{m}$  for a fitted feature
- Accepted to VDI/VDE 2634, not ISO 10360
- Weak on shiny, dark or translucent surfaces — dropout and noise

## CCD / CMOS machine vision

*A telecentric lens images the feature, usually back-lit in silhouette; sub-pixel edge detection locates the edge.*

- Sub-second per view — a full station fits inside a 90 s takt
- 50 mm field on a 5 MP sensor  $\approx$  20  $\mu\text{m}$  per pixel;  $\sim$ 10–30  $\mu\text{m}$  real accuracy
- Telecentric optics are essential — constant magnification with height
- Global-shutter CMOS specified; a rolling shutter distorts a moving part
- Two-dimensional, line of sight — blind bores are unreachable

# Head to head on the parameters that matter

| Parameter                 | Tactile CMM          | 3-D laser scanner     | CCD / CMOS vision           |
|---------------------------|----------------------|-----------------------|-----------------------------|
| Speed per block           | 25 min               | 2 – 5 min             | 45 s                        |
| Accuracy                  | 1.9 + L/300 µm       | 20 – 50 µm            | 10 – 30 µm                  |
| Repeatability             | 0.5 – 2 µm           | 5 – 20 µm             | 2 – 10 µm                   |
| Surface reflectivity      | Not affected         | Highly sensitive      | Moderately sensitive        |
| Coolant mist and dust     | Part must be clean   | Beam scatter, dropout | Needs sealed, purged optics |
| Vibration                 | Most sensitive       | Tolerant              | Most tolerant               |
| Blind / internal features | Reaches them         | Line of sight only    | Line of sight only          |
| GD&T capability           | Full                 | Form and profile      | Limited, 2-D                |
| Traceability standard     | ISO 10360 / IS 15250 | VDI/VDE 2634          | Gauge R&R only              |
| In-line capability        | Near-line at best    | Feasible on a robot   | Excellent                   |

# The arithmetic that settles it

A 1 500 s inspection cannot live inside a 90 s takt

**1 500 s**

full GD&T inspection  
on the CMM

**÷ 90 s**

line takt time

**17**

CMMs needed for  
100 % inspection

**6 %**

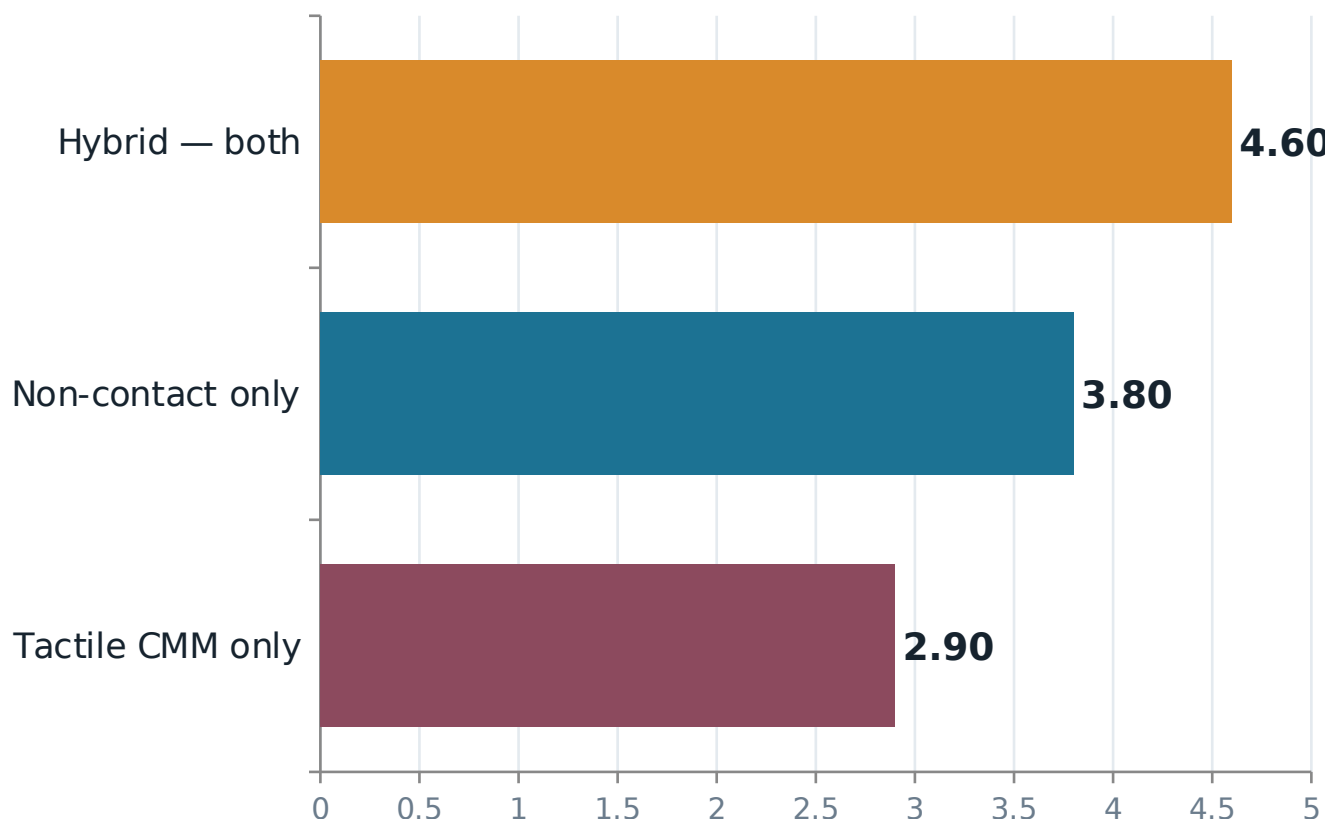
sampling one CMM  
can actually achieve

**In-line vision + laser:** 45 s < 90 s takt → 100 % of blocks gauged, every one

Seventeen CMMs is not an engineering proposal. A CMM can therefore only ever sample — and sampling one block in seventeen means a drift is found, on average, sixteen blocks too late. That is the very failure the plant is trying to eliminate.

# Weighted trade-off matrix

Seven criteria, weighted for this plant's problem — scores out of 5



## Criteria and weights

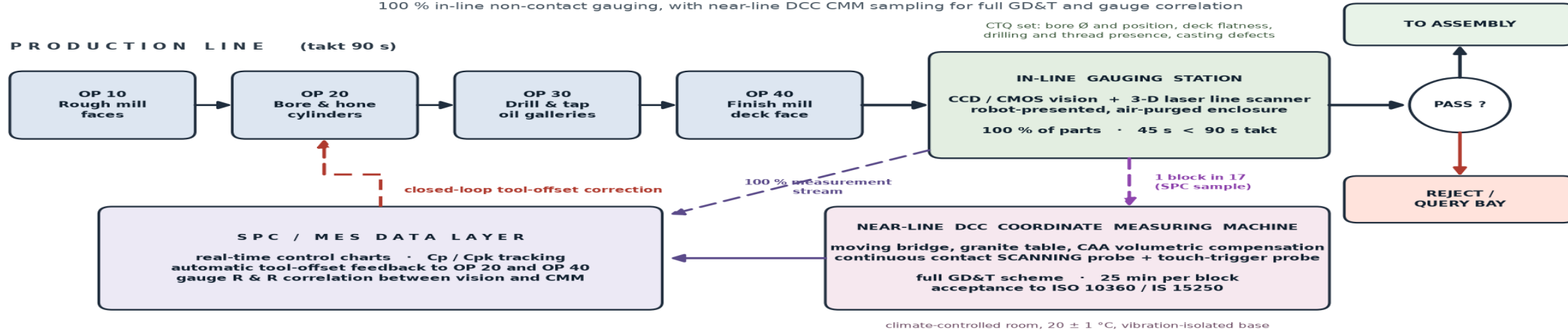
- 25 %** Cycle time and throughput
- 20 %** Accuracy on the CTQ features
- 15 %** GD&T and datum capability
- 15 %** In-line integration
- 10 %** Robustness to mist, dust, vibration
- 10 %** Capital and operating cost
- 5 %** SPC / MES data integration

# Recommendation: a hybrid architecture

100 % in-line non-contact gauging, backed by a near-line DCC scanning CMM sampling 1 block in 17

**Fig. A2.1 — Proposed hybrid inspection architecture for the cylinder-block line**

100 % in-line non-contact gauging, with near-line DCC CMM sampling for full GD&T and gauge correlation



## Why the hybrid, and not either technology on its own

- A full-GD&T CMM inspection takes 25 min against a 90 s takt — 17 machines would be needed for 100 % coverage, so a CMM alone can only ever sample.
- Non-contact gauging alone cannot reach blind bores, and cannot certify datum-referenced true position and cylindricity to a traceable standard.
- Together, vision catches every part and detects a process drift within one part, while the CMM certifies the measurement system and the full tolerance scheme.

## In-line station

100 % · 45 s · bore Ø and position, deck flatness, drilling and thread presence

## Near-line CMM

1 in 17 · 25 min · full GD&T to ISO 10360, and the reference for gauge R&R

## Closed loop

tool offsets fed back to OP 20 and OP 40 — a drift corrected within one part

# The business case, and what to do next

**₹ 1.35 cr**

annual saving from cutting  
scrap 2.0 % → 0.5 %

**₹ 1.30 cr**

total capital  
(vision ₹45 L + CMM ₹85 L)

**≈ 1.0 yr**

simple payback

*Illustrative figures — the structure of the case is what matters: the saving is driven by the scrap improvement and is an order of magnitude larger than the difference in capital between the options.*

## Two conditions for success

- Correlate the gauges formally — quarterly gauge R&R between the in-line station and the CMM, with in-line control limits guard-banded inside the drawing tolerance by the measured uncertainty.
- Protect the optics — a sealed, filtered, positive-pressure enclosure with an air knife. Most in-line vision installations that fail, fail for this reason, not for want of resolution.

**Thank you — questions welcome**